

NATHAN R. TATTRIE

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Senior Mechanical Engineer | Mechanism Design | Mechatronics | Robotics

Senior Mechanical Engineer with 7+ years leading electromechanical hardware from ambiguous requirements through concept development, detailed design, prototyping, design verification, supplier coordination, manufacturing readiness, and product launch. Background combines robotic mechanism design, mechatronics, and ruggedized medical/defense hardware at Roam Robotics with high-volume automotive production discipline from Ford's 2022 Expedition program.

TECHNICAL SKILLS

- Mechanical / Robotics: Electro-mechanical system ownership, robotics hardware, human-robot interaction, pneumatic actuation, mechanism design, linkage and joint design, compressor and motor debug, mechanism packaging, torque analysis, thermal management, acoustic reduction, ruggedization, root cause analysis
- CAD / Analysis / Quality: SolidWorks, advanced assemblies, FEA, PDM, Arena PLM, GD&T (ANSI Y14.5), tolerance analysis, DFMEA, DVP&R, design verification, validation planning, BOM management, ECO/DCO documentation, test data
- Manufacturing / Product Development: DFM/DFA, rapid prototyping, machined parts, CNC machining, mill, lathe, 3D printing, waterjet, laser cutting, casting, injection molding, stamping, extrusion, welding, softgoods integration, manufacturing readiness, product launch
- Program / Operations: Supplier coordination, interdisciplinary team collaboration, PPAP readiness, functional test procedures, calibration records, MPIs, end-of-line testing, failure analysis, quality assurance

EXPERIENCE

ROAM ROBOTICS | Senior NPI Mechanical Engineer, Medical & Defense | Feb 2021 - Present

- Own mechanical design and performance of wearable robotic electro-mechanical systems, advancing R&D prototypes into ruggedized defense hardware and Class I medical devices.
- Developed powered knee orthosis mechanisms with pneumatic actuation and precision mechanical subsystems, analyzing torque transfer across thigh, knee, and ankle interfaces for controls tuning, user fit, and assistive performance in a human-robot interactions.
- Designed and qualified adjustable torque-test fixtures for powered knee joints using machined components, CAD, free-body analysis, and force measurement to validate torque output across multiple joint angles.
- Redesigned Roam's powerpack from backpack-sized prototype to waist-worn system, reducing weight from 15.4 lb to 9.7 lb while preserving airflow performance, improving ergonomics, and enabling defense customer deployment.
- Led mechatronics and controls-facing compressor development by diagnosing leakage, redesigning reed valves, characterizing startup torque, and validating a pressure relief strategy, which improved airflow by 80% and reduced high-pressure stall behavior.
- Built design verification fixtures and functional test procedures for actuators, compressors, enclosures, remotes, electronics, and softgoods. Tested hundreds of specimens, performed life-cycle testing and failure analysis, and redesigned flawed subcomponents without missing milestones or compromising fit and function.
- Improved field robustness through sealed enclosures, conformal coatings, softgoods integration, thermal management, and acoustic design, which achieved IK08 impact and IP57 ingress protection while reducing system noise from 85 dBA to 60 dBA.
- Authored PRD, ERD, DFMEA, test matrix, MPIs, end-of-line tests, calibration records, and fixture IQ/OQ/PQ documentation while maintaining CAD models, drawings, test data, BOMs, ECOs, DCOs, and document control in Arena PLM.
- Partnered with manufacturing, quality, supply chain, and domestic/international vendors to source components, manage schedules, and release machined, extruded, injection molded, and softgoods components on aggressive timelines.
- Collected field data with Air Force users during routine activities and supported volunteer measurement, fit calibration, and fabrication inputs for custom medical devices.

FORD MOTOR COMPANY | Product Design Engineer, Exterior Lighting Team | Aug 2019 - Jan 2021

- Owned mechanical design and release of all exterior lighting systems for the 2022 Ford Expedition, including headlights, tail lamps, fog lamps, license plate lamps, CHMSL, and side mirror lamps. Approximately 62,000 units were sold in 2022.
- Led precision housing and internal component development from concept through validation, assembly, and product launch while coordinating interfaces to harnessing, sheet metal, fascia, grille, and tailgate teams.
- Partnered with studio and cross-functional vehicle teams to meet aesthetic, Class A surface, optical, and cosmetic fit-and-finish requirements across production lighting assemblies.
- Resolved out-of-spec body-to-lamp gaps by coordinating tolerance changes across lighting housings and adjacent fascia/sheet metal interfaces, improving fit and finish for high-volume production launch.
- Worked with interdisciplinary teams spanning electrical, optics, studio, vehicle integration, and suppliers to deliver cost-, weight-, power-, and package-efficient lamp assemblies for Expedition, autonomous vehicle, and F-150 Lightning programs.
- Integrated circuitry into production lamp designs while accounting for EMC, worst-case circuit analysis, LED and electronic thermal constraints, harness interfaces, environmental durability, and vehicle assembly requirements.
- Applied CAD, GD&T, DFMEA/FMA, CAE, DVP&R, and physical testing to resolve manufacturability, performance, tolerance, and lifecycle tradeoffs across production lighting systems.
- Partnered with Tier 1 suppliers in Asia and North America on tooling, DFM, validation strategy, PPAP readiness, production release, and supplier drawing review for production lighting components.

EDUCATION

University of Michigan, College of Engineering | B.S. Mechanical Engineering, Minor in Entrepreneurship | May 2019